

SEMTECH UNIFIED SOFTWARE PLATFORM (USP)

# Radio Test CLI User Manual

Interactive RF Test Interface for Semtech Radio Modules

SX126x • LR11xx • LR20xx

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**Version:** v0.5.1

**Platforms:** NUCLEO-L476RG • NUCLEO-L073RZ • FPB-RA0E2 • Linux (Raspberry Pi) • Zephyr (nRF54L15)

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PART 1

## Quick Start

Build the CLI, connect your radio, and run your first test.

## 1.1 Requirements

**Semtech radio module:** LR1110, LR1120, LR1121, LR2021, LR2022, SX1261, SX1262, or SX1268 connected via SPI/GPIO.

### Linux (Raspberry Pi)

| ITEM           | REQUIREMENT                                                       |
|----------------|-------------------------------------------------------------------|
| Host platform  | Raspberry Pi with SPI enabled                                     |
| SPI interface  | <code>/dev/spidev0.0</code>                                       |
| GPIO interface | <code>/dev/gpiochip0</code>                                       |
| User groups    | User must be in the <code>spi</code> and <code>gpio</code> groups |
| Build tools    | GCC 12+, CMake 3.25+, Ninja                                       |

### Baremetal (STM32 / RA0E2)

| ITEM             | REQUIREMENT                                                                 |
|------------------|-----------------------------------------------------------------------------|
| Evaluation board | NUCLEO-L476RG, NUCLEO-L073RZ, or FPB-RA0E2                                  |
| Build tools      | ARM GCC toolchain ( <code>arm-none-eabi-gcc</code> 12+), CMake 3.25+, Ninja |
| Serial terminal  | ANSI-capable emulator (e.g. Tera Term, PuTTY, minicom) at 115200 baud       |

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## 1.2 First Test: CW in Three Commands

---

```
region us
power 22
start cw --freq 915.0
```

The radio is now transmitting an unmodulated carrier at 915 MHz, +22 dBm. Type `stop` or press **Ctrl+C** to halt.

**What just happened:** `region us` loaded US915 defaults. `power 22` set the PA. `start cw` started transmission with a one-shot frequency override. No modulation setup is needed for CW.

The CLI saves history to `.radio_test_history`. Use **Up/Down arrows** to recall previous commands. Press **Tab** to complete commands and parameter values.

**PART 2**

## Test Procedures

Step-by-step guides for each test scenario — from CW power measurements to FCC pre-compliance and PER link tests. Each section explains purpose, parameter choices, and what to expect. For complete parameter syntax, see Part 3.

## 2.1 CW / Conducted Power Measurement

**Purpose:** Measure conducted output power and verify frequency accuracy. Also used to confirm PA configuration before modulated testing.

### Minimum setup

```
region us
power 22
start cw --freq 915.0
```

No modulation is required for CW. The radio transmits a pure carrier at the specified frequency and power.

### What to adjust

| PARAMETER                                  | HOW TO SET IT                                        | WHY                                                       |
|--------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| <code>freq</code>                          | Band edges and center (e.g. 902.3, 915.0, 927.5 MHz) | Verify power flatness across the band                     |
| <code>power</code>                         | Step through the power range                         | Confirm power accuracy at each level                      |
| <code>pa</code><br><code>pa_ramp_us</code> | <code>pa pa_ramp_us 16</code>                        | Faster ramp; observe switching transients on the analyzer |

### Inline power sweep (scripted)

```
region us
start cw --freq 915.0 --power 22
delay 10000
stop
start cw --freq 915.0 --power 14
delay 10000
stop
start cw --freq 915.0 --power 0
delay 10000
stop
exit
```

`--freq` and `--power` on `start` are one-shot overrides. They do not change the stored configuration, so you can sweep without reconfiguring between runs.

## 2.2 FCC FHSS Test — 64-Channel Hopping

**Purpose:** Demonstrate compliance with FCC §15.247(a)(1) — frequency hopping spread spectrum. The radio hops across all 64 US915 uplink channels at the configured data rate, proving the system uses at least 50 hopping frequencies with a channel occupancy time below 400 ms.

### Minimum setup

```
region us
modulation lora
start fhss --dr 0 --count 0
```

This is a complete, valid FHSS test. The CLI loads LoRa defaults for DR0 (SF10/125 kHz), computes the maximum compliant payload, and hops indefinitely through all 64 channels.

### Which DR to use

| DR  | SF   | BW      | NOTES                                                         |
|-----|------|---------|---------------------------------------------------------------|
| DR0 | SF10 | 125 kHz | Most commonly requested by test labs; longest airtime per hop |
| DR3 | SF7  | 125 kHz | Shortest airtime at 125 kHz; useful for rapid channel cycling |
| DR4 | SF8  | 500 kHz | 500 kHz bandwidth; different PA path on some chips            |

For FCC pre-compliance the test lab will specify which DR to use. DR0 is most commonly requested.

### Dwell time

At DR0 (SF10/125 kHz) with the maximum auto-computed payload, each hop occupies approximately 370 ms — safely under the FCC 400 ms limit. The CLI warns if a manual payload size would exceed this limit.

### Adding inter-hop delay

```
start fhss --dr 0 --count 64 --delay 100
```

The `--delay` option adds a pause (in milliseconds) between hops. This gives the spectrum analyzer time to settle and capture each channel cleanly. A delay of 50–100 ms is typical for swept-tuned analyzers.



## Verifying the channel plan

```
show channels
```

Lists all 64 US915 uplink channels with their index and frequency. Use this to verify the channel plan matches your test plan before starting.

## 2.3 FCC DTS Test — 500 kHz Digital Modulation

**Purpose:** Demonstrate compliance with FCC §15.247(a)(3) — digital transmission system (DTS). The radio transmits continuously on a single frequency using 500 kHz LoRa bandwidth, proving the modulated signal meets spectral mask and power requirements without frequency hopping.

### Setup

```
region us
modulation lora
bw 500
sf 7
cr 4/5
pld auto
start dts --freq 915.0
```

#### Why these parameters:

- `bw 500` — 500 kHz is required for DTS qualification under §15.247(a)(3)
- `sf 7` — lowest spreading factor gives the widest occupied bandwidth, stressing the spectral mask
- `cr 4/5` — minimum coding rate (LoRaWAN standard)
- `pld auto` — maximum payload; no dwell-time limit applies at 500 kHz

No dwell-time limit applies to 500 kHz DTS operation.

### Varying the test frequency

```
start dts --freq 903.0
start dts --freq 915.0
start dts --freq 927.0
```

## 2.4 FCC Hybrid Test — 8-Channel Sub-Band

**Purpose:** Demonstrate compliance for systems that hop across fewer than 50 channels within an 8-channel sub-band of the US915 plan. The radio hops through 8 consecutive channels using 500 kHz LoRa bandwidth.

### Setup

```
region us
modulation lora
bw 500
sf 7
start hybrid --mask 0 --dr 4 --count 0
```

### Sub-band mask reference

| MASK | CHANNELS | FREQUENCY RANGE |
|------|----------|-----------------|
| 0    | ch 0–7   | 902.3–903.7 MHz |
| 1    | ch 8–15  | 903.9–905.3 MHz |
| 2    | ch 16–23 | 905.5–906.9 MHz |
| 3    | ch 24–31 | 907.1–908.5 MHz |
| 4    | ch 32–39 | 908.7–910.1 MHz |
| 5    | ch 40–47 | 910.3–911.7 MHz |
| 6    | ch 48–55 | 911.9–913.3 MHz |
| 7    | ch 56–63 | 913.5–914.9 MHz |

DR4 (SF8/500 kHz) is the standard LoRaWAN data rate for 500 kHz channels and is the natural choice for hybrid testing.

## Sweeping all sub-bands (script)

```
start hybrid --mask 0 --dr 4 --count 8 --delay 100
start hybrid --mask 1 --dr 4 --count 8 --delay 100
start hybrid --mask 2 --dr 4 --count 8 --delay 100
start hybrid --mask 3 --dr 4 --count 8 --delay 100
start hybrid --mask 4 --dr 4 --count 8 --delay 100
start hybrid --mask 5 --dr 4 --count 8 --delay 100
start hybrid --mask 6 --dr 4 --count 8 --delay 100
start hybrid --mask 7 --dr 4 --count 8 --delay 100
```

## 2.5 PER Link Test

**Purpose:** Measure packet error rate between two radio units to verify link budget, sensitivity, and antenna performance. One unit transmits a known number of packets; the other counts received packets and reports RSSI, SNR, and PER.

### Setup — Unit A (transmitter)

```
region us
modulation lora
freq 915.0
power 14
bw 125
sf 10
cr 4/5
per count 1000
per interval 200
per payload 16
start per-tx
```

### Setup — Unit B (receiver)

Configure identically (region, modulation, freq, bw, sf, cr, syncword, header, crc), then:

```
region us
modulation lora
freq 915.0
bw 125
sf 10
cr 4/5
start per-rx
```

Start the receiver before the transmitter. The receiver must be listening before the first packet is sent, or the initial packets will be lost and inflate the PER result.

### Reading results

```
per stats
```

```
Last PER test result (rx):
```

```
Received: 997
```

```
CRC errors: 0
```

```
PER: 0.30%
```

```
RSSI avg: -78 dBm
```

```
SNR avg: +9.2 dB
```

## Sensitivity sweep

Reduce TX power in steps to find the sensitivity threshold where PER degrades:

```
start per-tx --power 14
```

```
start per-tx --power 10
```

```
start per-tx --power 5
```

```
start per-tx --power 0
```

```
start per-tx --power -5
```

## 2.6 FLRC Test (LR2021)

**Purpose:** Test the FLRC (Fast Long Range Communication) modulation mode available on the LR2021 radio. FLRC provides higher data rates (up to 2600 kbps). Available on LR2021 only, in `us` and `2g4` regions.

### Conducted power / spectral mask

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
header variable
crc 2
pld 127
start modulated --freq 2440.0 --power 10
```

#### Parameter guidance:

- `bt bt0.5` — Gaussian BT product of 0.5 gives a good balance of spectral efficiency and ISI; tighter spectral mask
- `bt off` — disables Gaussian filtering for maximum data throughput at the cost of wider spectral occupancy
- `cr none` — no coding rate overhead; use `1/2` or `3/4` for forward error correction in noisy environments

### FLRC PER link test

#### Unit A (TX):

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
syncword ED592398
header variable
crc 2
per count 1000
per interval 50
per payload 64
start per-tx --freq 2440.0 --power 10
```

### Unit B (RX):

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
syncword ED592398
header variable
crc 2
start per-rx --freq 2440.0
```

Both units must use identical `br`, `cr`, `bt`, `preamble`, `sw_len`, `syncword`, `header`, and `crc` settings. The `syncword` value must match between transmitter and receiver.



**PART 3**

# Command Reference

Complete syntax and valid values for every command and parameter. Use this when you need exact details — Part 2 covers when and why to use them.

## 3.1 Region and Modulation

### `region` — Set Regulatory Region

```
region <us|2g4>
```

Sets the regulatory region and loads default frequency and power values. Must be the first command issued after startup.

| REGION ID        | BAND   | FREQUENCY RANGE   | MAX POWER | REGULATORY MODES  |
|------------------|--------|-------------------|-----------|-------------------|
| <code>us</code>  | US915  | 902.0–928.0 MHz   | +22 dBm   | FHSS, DTS, Hybrid |
| <code>2g4</code> | WW-2G4 | 2400.0–2480.0 MHz | +12 dBm   | —                 |

### `modulation` — Set Modulation Type

```
modulation <lora|flrc>  
modulation help
```

Sets the modulation type and loads region-appropriate parameter defaults. `modulation help` lists all modulations supported by the connected chip.

LoRa is available on all chips. FLRC is available on LR2021 only, in `us` and `2g4` regions.

### `freq` — Set Center Frequency

```
freq <MHz>
```

Sets the transmit/receive center frequency in MHz. The value must be within the current region's allowed range.

### `power` — Set TX Power

```
power <dBm>
```

Sets the transmit output power in dBm. The value is validated against the current region's minimum and maximum limits.

**agc — Receiver Gain Control**

```
agc <auto|g1|g2|...|g13>
```

Sets the receiver gain mode. `auto` (default) enables automatic gain control. `g1` through `g13` select a fixed gain step (g1 = maximum gain, g13 = minimum gain). Available on LR20xx chips only.

| VALUE                 | DESCRIPTION                                     |
|-----------------------|-------------------------------------------------|
| <code>auto</code>     | Automatic gain control (default)                |
| <code>g1 - g13</code> | Fixed gain step (g1 = max gain, g13 = min gain) |

**boost-lf / boost-hf — RX Boost Level**

```
boost-lf <auto|0-7>
boost-hf <auto|0-7>
```

Sets the RX boost level for the low-frequency (`boost-lf`) or high-frequency (`boost-hf`) receive path. `auto` uses the default value (0 for LF, 4 for HF). Available on LR20xx chips only.

**xosc — Crystal Oscillator Trimming**

```
xosc <show|default|xta|xtb[|wait]> [value]
```

Manages crystal oscillator trimming capacitor values. Available on LR20xx and SX126x chips.

| SUBCOMMAND                           | CHIPS       | DESCRIPTION                                             |
|--------------------------------------|-------------|---------------------------------------------------------|
| <code>xosc show</code>               | All         | Display current XTA and XTB values (with pF conversion) |
| <code>xosc default</code>            | All         | Restore BSP default trim values                         |
| <code>xosc xta &lt;0-47&gt;</code>   | All         | Set XTA capacitor trim ( $11.3 + N \times 0.47$ pF)     |
| <code>xosc xtb &lt;0-47&gt;</code>   | All         | Set XTB capacitor trim ( $11.1 + N \times 0.47$ pF)     |
| <code>xosc wait &lt;0-255&gt;</code> | LR20xx only | Set stabilization delay in microseconds                 |

On SX126x the XOSC stabilisation delay is managed by the hardware state machine and is not user-configurable; `xosc wait` is not available. Values set via `xosc xta` / `xosc xtb` persist across CLI commands and are re-applied automatically before every TX/RX operation.

**status — Show Current Configuration**`status`

Displays all current configuration parameters in a single view, including region, modulation, frequency, power, modulation-specific parameters, PA overrides, and test mode state.

## 3.2 LoRa Parameters

LoRa parameters are set by typing the parameter name directly at the prompt after `modulation lora` is active. All parameters are validated against regional constraints before being applied.

### `bw` — Bandwidth

```
bw <kHz>
```

| REGION | VALID VALUES (KHZ) |
|--------|--------------------|
| US915  | 125, 250, 500      |
| WW-2G4 | 812                |

Changing bandwidth may invalidate the current spreading factor. The CLI warns when this occurs; use `sf` to adjust before starting a test.

### `sf` — Spreading Factor

```
sf <5-12>
```

| BW (KHZ) | US915 SF RANGE | LORAWAN DR EQUIVALENT  |
|----------|----------------|------------------------|
| 125      | SF5–SF10       | DR0 (SF10) – DR3 (SF7) |
| 250      | SF5–SF12       | —                      |
| 500      | SF5–SF12       | DR4 (SF8/500k)         |

### `cr` — Coding Rate

```
cr <4/5|4/6|4/7|4/8> – all chips
cr <li4/5|li4/6|li4/8> – LR11xx, LR20xx only
cr <lic4/6|lic4/8> – LR20xx only
```

Sets the LoRa coding rate. Standard rates `4/5` through `4/8` are available on all chips. Long interleaved (LI) rates are available on LR11xx and LR20xx. Long interleaved convolutional (LI-Conv) rates are LR20xx only.

| VALUE               | TYPE                           | CHIPS          |
|---------------------|--------------------------------|----------------|
| 4/5, 4/6, 4/7, 4/8  | Standard                       | All            |
| li4/5, li4/6, li4/8 | Long interleaved               | LR11xx, LR20xx |
| lic4/6, lic4/8      | Long interleaved convolutional | LR20xx only    |

### `preamble` — Preamble Length

```
preamble <4-65535>
```

Sets the LoRa preamble length in symbols. Default is 8 symbols (LoRaWAN standard). Longer preambles improve reception reliability at the cost of airtime.

### `syncword` — Sync Word

```
syncword <hex>
```

Sets the LoRa sync word as a one-byte hexadecimal value.

| VALUE | MEANING                                 |
|-------|-----------------------------------------|
| 0x34  | LoRaWAN public network (default for US) |
| 0x12  | Private network                         |
| 0x21  | WW-2G4 public (default for 2g4)         |

### `header` — Header Mode

```
header <explicit|implicit>
```

In **explicit** mode (default), the packet header carries the payload length, coding rate, and CRC presence flag. In **implicit** mode, no header is transmitted — both TX and RX must agree on the payload length and CRC setting in advance.

### `crc` — CRC Enable

```
crc <on|off>
```

Enables or disables the LoRa payload CRC. Default is `on`.

**iq — I/Q Polarity**

```
iq <standard|inverted>
```

Sets the I/Q polarity. `standard` (default) uses normal I/Q mapping. `inverted` swaps I and Q, which is used by some LoRaWAN downlink configurations. Available on all chips.

**pld — Payload Size**

```
pld <1-255|auto>
```

Sets the payload size in bytes for continuous modulated transmission ( `start modulated` ). Setting `auto` computes the maximum payload that fits within the FCC 400 ms dwell-time limit for the current LoRa configuration.

**FCC dwell time:** At 125 kHz and 250 kHz bandwidths, the FCC limits each transmission to 400 ms per channel before hopping is required. The CLI calculates time-on-air and warns if the configured payload exceeds this limit. 500 kHz (DTS) is exempt from the dwell-time rule.

**ldro — Low Data Rate Optimization**

```
ldro <on|off|auto>
```

Controls the low data rate optimization flag. When set to `auto` (default), the CLI enables LDRO automatically when the symbol duration exceeds 16 ms (SF11/125 kHz and SF12/125 kHz). Force `on` or `off` to override. Available on SX126x and LR11xx only (not available on LR20xx).

### 3.3 FLRC Parameters

FLRC parameters are available after `modulation flrc` is active. FLRC is supported on LR2021 only, in `us` and `2g4` regions.

#### `br` — Bit Rate

```
br <260|325|520|650|1040|1300|2080|2600>
```

Sets the FLRC bit rate in kbps. Higher bit rates reduce airtime but require better link budget.

| VALUE (KBPS) | BANDWIDTH |
|--------------|-----------|
| 260          | 312 kHz   |
| 325          | 312 kHz   |
| 520          | 624 kHz   |
| 650          | 624 kHz   |
| 1040         | 1248 kHz  |
| 1300         | 1248 kHz  |
| 2080         | 2496 kHz  |
| 2600         | 2496 kHz  |

#### `cr` — Coding Rate

```
cr <1/2|2/3|3/4|none>
```

Sets the FLRC forward error correction coding rate. `none` disables FEC for maximum throughput. Default is `3/4`.

#### `bt` — Gaussian BT Filter

```
bt <off|bt0.5|bt1>
```

Controls the Gaussian pulse-shaping filter. `bt0.5` gives tighter spectral mask (recommended). `bt1` provides moderate filtering. `off` disables filtering for maximum throughput. Default is `bt0.5`.



**preamble** — Preamble Length

```
preamble <4|8|12|16|20|24|28|32>
```

Sets the FLRC preamble length in bits. Must be a multiple of 4 from 4 to 32.

**sw\_len** — Sync Word Length

```
sw_len <off|2|4>
```

Sets the sync word length in bytes. `off` disables the sync word. Default is `4`.

**tx\_sw** — TX Sync Word Index

```
tx_sw <off|1|2|3>
```

Selects which syncword register to transmit with. `off` disables syncword transmission. Default is `1`.

**rx\_sw** — RX Sync Word Match

```
rx_sw <off|1-7>
```

Selects which syncword register(s) to match on receive, as a bitmask. Default is `1` (syncword 1 only).

| VALUE                             | MATCHES                           |
|-----------------------------------|-----------------------------------|
| <code>off</code> / <code>0</code> | None (syncword matching disabled) |
| <code>1</code>                    | sw1                               |
| <code>2</code>                    | sw2                               |
| <code>3</code>                    | sw1   sw2                         |
| <code>4</code>                    | sw3                               |
| <code>5</code>                    | sw1   sw3                         |
| <code>6</code>                    | sw2   sw3                         |
| <code>7</code>                    | sw1   sw2   sw3                   |

**header** — Header Mode

```
header <variable|fixed>
```

**variable** (default) includes the payload length in the packet header. **fixed** omits the length header — both TX and RX must agree on the payload length.

**crc** — CRC Length

```
crc <off|2|3|4>
```

Sets the CRC length in bytes. **off** disables CRC. Default is **2**.

**syncword** — Sync Word Value

```
syncword <hex>
```

Sets the 4-byte syncword value for register 1. Enter 1 to 8 hex digits; values shorter than 8 digits are zero-padded from the left. Default is **0xED592398**.

**FLRC syncword architecture:** **syncword** sets the byte values in the syncword register. **tx\_sw** selects which register to transmit with. **rx\_sw** controls which registers are matched on receive. For basic operation, set **syncword** to the desired value and leave **tx\_sw** and **rx\_sw** at their defaults (both **1**).

## 3.4 PA Configuration

The power amplifier is automatically configured when the `power` command is used. For advanced measurements requiring manual PA control, use the `pa` command.

```
pa show
pa reset
pa <param> <value>
```

`pa show` displays the current PA register values and valid ranges. `pa reset` clears all manual overrides and restores the automatic PA configuration for the current power level. `pa <param> <value>` sets an individual PA register.

### Override semantics:

- Manual PA overrides take effect immediately and persist until `pa reset` is called or the `power` command is used
- The `power` command recalculates all PA registers from the target power level, clearing any manual overrides
- Overrides are not saved between sessions

### LR20xx (LR2021, LR2022)

| PARAMETER                     | RANGE           | DESCRIPTION                           |
|-------------------------------|-----------------|---------------------------------------|
| <code>half_power</code>       | –39 to 44       | TX power in 0.5 dB steps              |
| <code>pa_sel</code>           | 0–1 (read-only) | PA path: 0 = LF, 1 = HF               |
| <code>pa_lf_duty_cycle</code> | 0–31            | Low-freq PA duty cycle                |
| <code>pa_lf_slices</code>     | 0–7             | Low-freq PA number of slices          |
| <code>pa_hf_duty_cycle</code> | 16–31           | HF duty cycle (16 = max, 31 = lowest) |
| <code>pa_ramp_us</code>       | 2–304           | PA ramp time in microseconds          |

`pa_sel` is read-only and set automatically based on frequency: LF PA for sub-GHz, HF PA for 2.4 GHz. `pa_ramp_us` accepts discrete values only: 2, 4, 8, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 240, 272, 304.

**SX1262 / SX1268**

| PARAMETER                  | RANGE    | DESCRIPTION                             |
|----------------------------|----------|-----------------------------------------|
| <code>power</code>         | −9 to 22 | TX power register value (dBm)           |
| <code>pa_duty_cycle</code> | 0–4      | PA duty cycle (max 0x04, see datasheet) |
| <code>hp_max</code>        | 0–7      | HP PA limit (7 = max, enables +22 dBm)  |
| <code>pa_ramp_us</code>    | 10–3400  | PA ramp time in microseconds            |

**SX1261**

| PARAMETER                  | RANGE     | DESCRIPTION                           |
|----------------------------|-----------|---------------------------------------|
| <code>power</code>         | −17 to 14 | TX power register value (dBm)         |
| <code>pa_duty_cycle</code> | 0–7       | PA duty cycle (see datasheet for max) |
| <code>pa_ramp_us</code>    | 10–3400   | PA ramp time in microseconds          |

**LR11xx (LR1110, LR1120, LR1121)**

| PARAMETER                  | RANGE           | DESCRIPTION                     |
|----------------------------|-----------------|---------------------------------|
| <code>power</code>         | −17 to 22       | TX power in dBm                 |
| <code>pa_sel</code>        | 0–2 (read-only) | PA path: 0 = LP, 1 = HP, 2 = HF |
| <code>pa_reg_supply</code> | 0–1 (read-only) | PA supply: 0 = VREG, 1 = VBAT   |
| <code>pa_duty_cycle</code> | 0–7             | PA duty cycle                   |
| <code>pa_hp_sel</code>     | 0–7             | HP PA slices                    |
| <code>pa_ramp_us</code>    | 16–304          | PA ramp time in microseconds    |

`pa_sel` and `pa_reg_supply` are read-only — the BSP selects LP/HP/HF based on frequency and power level. `pa_ramp_us` accepts discrete values only: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 240, 272, 304.

## 3.5 Test Modes and Inline Overrides

Test modes are started with `start <mode>` and stopped with `stop` or **Ctrl+C**. Only one mode may be active at a time. Configuration commands are blocked while any mode is running.

On baremetal MCU targets, the CLI runs in a polled main loop. `stop` is the primary way to halt a test. **Ctrl+C** is handled on Linux only.

### `start cw` — Unmodulated Carrier

Transmits a continuous single-frequency unmodulated carrier. No modulation needs to be configured — only frequency and power are required.

```
start cw [--freq <MHz>] [--power <dBm>]
```

### `start modulated` — Continuous Modulated Burst

Transmits packets back-to-back using the current modulation configuration. Used for occupied bandwidth measurements and spectral mask tests. Payload size is set with the `pld` command.

```
start modulated [--freq <MHz>] [--power <dBm>]
```

### `start rx` — Continuous Receive

Places the radio in continuous receive mode. Each received packet is printed with RSSI and SNR. CRC errors are reported separately.

```
start rx [--freq <MHz>]
```

### `start fhss` — Frequency Hopping (US only)

Transmits using the US915 64-channel FHSS plan. FCC §15.247(a)(1) pre-compliance.

```
start fhss [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

### `start dts` — Digital Transmission System (US only)

Transmits continuously on a fixed frequency at 500 kHz bandwidth. FCC §15.247(a)(3) pre-compliance.

```
start dts [--freq <MHz>] [--power <dBm>] [--count <N>] [--delay <ms>]
```

### `start hybrid` — 8-Channel Sub-Band (US only)

Hops through 8 consecutive channels within a selectable sub-band. The `--mask` argument is required.

```
start hybrid --mask <0-7> [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

### `start per-tx` / `start per-rx` — Packet Error Rate

Runs a PER test between two radio units. Configure PER parameters with the `per` command before starting.

```
start per-tx [--freq <MHz>] [--power <dBm>]
start per-rx [--freq <MHz>]
```

### `stop` — Stop Active Test

Stops the currently running test mode and returns to idle.

```
stop
```

## Inline overrides

The `start` command accepts inline overrides that modify parameters for a single test run without changing the stored configuration.

| OVERRIDE                         | APPLICABLE MODES  | DESCRIPTION                     |
|----------------------------------|-------------------|---------------------------------|
| <code>--freq &lt;MHz&gt;</code>  | All               | One-shot frequency override     |
| <code>--power &lt;dBm&gt;</code> | All TX modes      | One-shot power override         |
| <code>--dr &lt;0-6&gt;</code>    | fhss, hybrid      | LoRaWAN data rate for this run  |
| <code>--count &lt;N&gt;</code>   | fhss, dts, hybrid | Packet/hop count (0 = infinite) |
| <code>--delay &lt;ms&gt;</code>  | fhss, dts, hybrid | Inter-packet/hop delay          |
| <code>--mask &lt;0-7&gt;</code>  | hybrid (required) | 8-channel sub-band selection    |

## 3.6 PER Configuration

| COMMAND                                     | DESCRIPTION                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------------------|
| <code>per count &lt;N&gt;</code>            | Packets TX will send; 0 = run indefinitely (default: 100)                           |
| <code>per interval &lt;ms&gt;</code>        | Inter-packet gap on TX side; must be > 0 (default: 500 ms)                          |
| <code>per payload &lt;6-255 auto&gt;</code> | Payload size per packet; <code>auto</code> = max for FCC dwell time (default: auto) |
| <code>per stats</code>                      | Display last test result (available any time, even during a test)                   |
| <code>per reset</code>                      | Clear last saved result                                                             |

`per stats` and `per reset` are always available. All other `per` subcommands require the radio to be idle — use `stop` first.

## 3.7 Utility Commands and Scripting

### `show channels`

Displays the complete LoRaWAN channel plan for the current region, listing each channel index and its center frequency.

```
show channels
```

### `delay`

Pauses execution for the specified duration in milliseconds. If a test mode is active, it continues to run during the delay — this is intended for use in scripts to set observation windows.

```
delay <ms>
```

### `term` — Terminal Input Mode

```
term <ansi|plain>
```

Switches the terminal input mode between `ansi` (arrow keys, history, tab completion) and `plain` (basic line input). MCU only — not available on Linux. With no arguments, shows the current mode.

### `version`

Displays the CLI version string and build information.

```
version
```

### `help`

Shows the command reference. Provide a command name for context-sensitive detail.

```
help [command]
```

### `exit` / `quit`

Stops any active test, closes the radio, and exits the CLI.



```
exit
```

## Scripting / Pipe Mode

When stdin is not a terminal (i.e. piped or redirected input), the CLI runs in non-interactive script mode: commands are read sequentially from stdin and the CLI exits when EOF is reached.

```
./build/radio_test_cli <<'EOF'  
region us  
modulation lora  
freq 915.0  
power 22  
start cw  
delay 30000  
stop  
exit  
EOF
```

Commands that return an error do not stop script execution. Output is written to stdout in the same format as interactive mode, making it suitable for capture with `tee` or redirection.

**PART 4**

# Reference Tables

Region parameters, chip feature matrix, and the complete command quick-reference.

## 4.1 Region Reference

### US915

| PARAMETER             | VALUE                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------|
| Frequency range       | 902.0–928.0 MHz                                                                         |
| Default frequency     | 902.3 MHz (LoRaWAN ch0)                                                                 |
| TX power range        | –10 to +22 dBm (LR20xx); –9 to +22 dBm (SX1262/SX1268); –17 to +22 dBm (SX1261, LR11xx) |
| Default TX power      | +22 dBm                                                                                 |
| LoRa BW options       | 125, 250, 500 kHz                                                                       |
| LoRa SF (125 kHz)     | SF5–SF10                                                                                |
| LoRa SF (250/500 kHz) | SF5–SF12                                                                                |
| Default SF/BW         | SF10 / 125 kHz (DR0)                                                                    |
| Default sync word     | 0x34 (LoRaWAN public)                                                                   |
| Regulatory modes      | FHSS (64-ch), DTS (500 kHz), Hybrid (8-ch sub-band)                                     |

### US915 Data Rates

| DR  | MODULATION | SF   | BW      | BIT RATE (APPROX) |
|-----|------------|------|---------|-------------------|
| DR0 | LoRa       | SF10 | 125 kHz | 980 bps           |
| DR1 | LoRa       | SF9  | 125 kHz | 1760 bps          |
| DR2 | LoRa       | SF8  | 125 kHz | 3125 bps          |
| DR3 | LoRa       | SF7  | 125 kHz | 5470 bps          |
| DR4 | LoRa       | SF8  | 500 kHz | 12500 bps         |

### WW-2G4 (LR2021, LR2022, LR1120, LR1121)

| PARAMETER         | VALUE             |
|-------------------|-------------------|
| Frequency range   | 2400.0–2480.0 MHz |
| Default frequency | 2423.0 MHz        |

| PARAMETER         | VALUE                                            |
|-------------------|--------------------------------------------------|
| TX power range    | −17 to +12 dBm (LR20xx); −18 to +12 dBm (LR11xx) |
| Default TX power  | +10 dBm                                          |
| LoRa BW           | 812 kHz                                          |
| LoRa SF range     | SF5–SF12                                         |
| Default sync word | 0x21 (WW-2G4 public)                             |
| FLRC              | Available (LR2021 only)                          |
| Regulatory modes  | None (handled by host protocol stack)            |

## 4.2 Chip Compatibility

| FEATURE             | LR2021 | LR2022 | LR1110 | LR1120 | LR1121 | SX1261 | SX1262 | SX1268 |
|---------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| LoRa (Sub-GHz)      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| LoRa (2.4 GHz)      | ✓      | ✓      | —      | ✓      | ✓      | —      | —      | —      |
| FLRC                | ✓      | —      | —      | —      | —      | —      | —      | —      |
| Region: us          | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| Region: 2g4         | ✓      | ✓      | —      | ✓      | ✓      | —      | —      | —      |
| FHSS / DTS / Hybrid | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| PER TX/RX           | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| AGC control         | ✓      | ✓      | —      | —      | —      | —      | —      | —      |
| RX boost (LF/HF)    | ✓      | ✓      | —      | —      | —      | —      | —      | —      |
| XOSC trimming       | ✓      | ✓      | —      | —      | —      | ✓      | ✓      | ✓      |
| LED indicators      | —      | —      | ✓      | ✓      | ✓      | —      | —      | —      |
| I/Q polarity        | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| LDRO control        | —      | —      | ✓      | ✓      | ✓      | ✓      | ✓      | ✓      |
| CR LI/LI-Conv       | ✓      | ✓      | ✓      | ✓      | ✓      | —      | —      | —      |

## 4.3 Quick-Reference Command Table

| COMMAND                 | SYNTAX                                                                  | DESCRIPTION                                                                  |
|-------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <code>region</code>     | <code>region &lt;us 2g4&gt;</code>                                      | Set region; load defaults                                                    |
| <code>modulation</code> | <code>modulation &lt;lora flrc&gt;</code>                               | Set modulation; load defaults                                                |
| <code>freq</code>       | <code>freq &lt;MHz&gt;</code>                                           | Set center frequency                                                         |
| <code>power</code>      | <code>power &lt;dBm&gt;</code>                                          | Set TX power                                                                 |
| <code>agc</code>        | <code>agc &lt;auto g1-g13&gt;</code>                                    | Receiver gain control (LR20xx only)                                          |
| <code>boost-lf</code>   | <code>boost-lf &lt;auto 0-7&gt;</code>                                  | RX boost level, LF path (LR20xx only)                                        |
| <code>boost-hf</code>   | <code>boost-hf &lt;auto 0-7&gt;</code>                                  | RX boost level, HF path (LR20xx only)                                        |
| <code>xosc</code>       | <code>xosc &lt;show default xta xtb[ wait]&gt;</code>                   | Crystal oscillator trimming (LR20xx + SX126x; <code>wait</code> LR20xx only) |
| <code>status</code>     | <code>status</code>                                                     | Show complete current configuration                                          |
| <code>bw</code>         | <code>bw &lt;kHz&gt;</code>                                             | LoRa bandwidth (125/250/500/812)                                             |
| <code>sf</code>         | <code>sf &lt;5-12&gt;</code>                                            | LoRa spreading factor                                                        |
| <code>cr</code>         | <code>cr &lt;4/5 4/6 4/7 4/8 li4/5 li4/6 li4/8 lic4/6 lic4/8&gt;</code> | LoRa coding rate                                                             |
| <code>preamble</code>   | <code>preamble &lt;symbols&gt;</code>                                   | Preamble length                                                              |
| <code>syncword</code>   | <code>syncword &lt;hex&gt;</code>                                       | LoRa sync word (e.g. 0x34)                                                   |
| <code>header</code>     | <code>header &lt;explicit implicit&gt;</code>                           | LoRa header mode                                                             |
| <code>crc</code>        | <code>crc &lt;on off&gt;</code>                                         | LoRa CRC enable                                                              |
| <code>iq</code>         | <code>iq &lt;standard inverted&gt;</code>                               | LoRa I/Q polarity                                                            |
| <code>pld</code>        | <code>pld &lt;1-255 auto&gt;</code>                                     | Modulated TX payload size                                                    |
| <code>ldro</code>       | <code>ldro &lt;on off auto&gt;</code>                                   | Low data rate optimization (SX126x, LR11xx only)                             |
| <code>br</code>         | <code>br &lt;260 325 520 650 1040 1300 2080 2600&gt;</code>             | FLRC bit rate (kbps)                                                         |
| <code>cr</code> (FLRC)  | <code>cr &lt;1/2 2/3 3/4 none&gt;</code>                                | FLRC coding rate                                                             |
| <code>bt</code>         | <code>bt &lt;off bt0.5 bt1&gt;</code>                                   | FLRC Gaussian BT filter                                                      |

| COMMAND                             | SYNTAX                                              | DESCRIPTION                          |
|-------------------------------------|-----------------------------------------------------|--------------------------------------|
| <code>preamble</code><br>(FLRC)     | <code>preamble &lt;4 8 12 16 20 24 28 32&gt;</code> | FLRC preamble length (bits)          |
| <code>sw_len</code>                 | <code>sw_len &lt;off 2 4&gt;</code>                 | FLRC sync word length (bytes)        |
| <code>tx_sw</code>                  | <code>tx_sw &lt;off 1 2 3&gt;</code>                | FLRC TX sync word index              |
| <code>rx_sw</code>                  | <code>rx_sw &lt;off 1-7&gt;</code>                  | FLRC RX sync word match (bitmask)    |
| <code>header</code> (FLRC)          | <code>header &lt;variable fixed&gt;</code>          | FLRC header type                     |
| <code>crc</code> (FLRC)             | <code>crc &lt;off 2 3 4&gt;</code>                  | FLRC CRC length (bytes)              |
| <code>syncword</code><br>(FLRC)     | <code>syncword &lt;hex&gt;</code>                   | FLRC sync word value (e.g. ED592398) |
| <code>pa show</code>                | <code>pa show</code>                                | Display PA parameters and ranges     |
| <code>pa reset</code>               | <code>pa reset</code>                               | Clear PA overrides                   |
| <code>pa &lt;p&gt; &lt;v&gt;</code> | <code>pa &lt;param&gt; &lt;value&gt;</code>         | Set PA parameter manually            |
| <code>start cw</code>               | <code>start cw [overrides]</code>                   | Unmodulated carrier TX               |
| <code>start modulated</code>        | <code>start modulated [overrides]</code>            | Continuous modulated TX              |
| <code>start rx</code>               | <code>start rx [overrides]</code>                   | Continuous receive mode              |
| <code>start fhss</code>             | <code>start fhss [overrides]</code>                 | 64-channel FHSS TX (US only)         |
| <code>start dts</code>              | <code>start dts [overrides]</code>                  | DTS 500 kHz TX (US only)             |
| <code>start hybrid</code>           | <code>start hybrid --mask N [overrides]</code>      | Hybrid 8-ch TX (US only)             |
| <code>start per-tx</code>           | <code>start per-tx [overrides]</code>               | PER test transmitter                 |
| <code>start per-rx</code>           | <code>start per-rx [overrides]</code>               | PER test receiver                    |
| <code>stop</code>                   | <code>stop</code>                                   | Stop active test mode                |
| <code>per count</code>              | <code>per count &lt;N&gt;</code>                    | PER packet count (0 = infinite)      |
| <code>per interval</code>           | <code>per interval &lt;ms&gt;</code>                | PER inter-packet gap                 |
| <code>per payload</code>            | <code>per payload &lt;6-255 auto&gt;</code>         | PER payload size                     |
| <code>per stats</code>              | <code>per stats</code>                              | Show last PER test result            |
| <code>per reset</code>              | <code>per reset</code>                              | Clear last PER result                |
| <code>show channels</code>          | <code>show channels</code>                          | Display region channel plan          |

| COMMAND                               | SYNTAX                               | DESCRIPTION                    |
|---------------------------------------|--------------------------------------|--------------------------------|
| <code>delay</code>                    | <code>delay &lt;ms&gt;</code>        | Pause execution (scripting)    |
| <code>term</code>                     | <code>term &lt;ansi plain&gt;</code> | Terminal input mode (MCU only) |
| <code>version</code>                  | <code>version</code>                 | Show CLI version               |
| <code>help</code>                     | <code>help [command]</code>          | Show help                      |
| <code>exit</code> / <code>quit</code> | <code>exit</code>                    | Exit CLI                       |



**PART 5**

# Troubleshooting

Common errors and how to resolve them.

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## Troubleshooting

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### SPI device not found

**Symptom:** Failed to open `/dev/spidev0.0` on startup.

**Fix:** Enable SPI on the Raspberry Pi. Run `sudo raspi-config`, navigate to Interface Options > SPI, and enable it. Reboot. Verify with `ls /dev/spidev*`.

### Permission denied on SPI or GPIO

**Symptom:** Permission denied when accessing `/dev/spidev0.0` or `/dev/gpiochip0`.

**Fix:** Add your user to the required groups:

```
sudo usermod -aG spi,gpio $USER
```

Log out and back in for the group changes to take effect.

### Build fails on GCC 14

**Symptom:** Compilation error about incompatible pointer types in vendor code.

**Fix:** Add the warning suppression flag:

```
env CFLAGS="-Wno-incompatible-pointer-types" \  
  cmake -UCMAKE_C_FLAGS -S examples -B build -DBOARD=LINUX -DRAC_RADIO=lr2021 -G Ninja
```

### Radio not responding / no RX packets

**Symptom:** Commands succeed but no signal is observed on the spectrum analyzer, or the receiver sees no packets.

**Fix:** Check the following:

- Verify the SPI connection and that the correct radio shield is seated properly
- Confirm the `-DRAC_RADIO=` build flag matches the actual radio chip connected
- For PER tests, ensure both units have identical modulation parameters (bw, sf, cr, syncword, header, crc)
- Check that the frequency is within the selected region's valid range

---

## FLRC: "not supported by this chip"

**Symptom:** `modulation flrc` returns an error.

**Fix:** FLRC is available on LR2021 only. Verify the build was compiled with `-DRAC_RADIO=lr2021`. FLRC is available in both `us` and `2g4` regions.

## Region 2g4: "not supported by this chip"

**Symptom:** `region 2g4` returns an error on SX126x or LR11xx.

**Fix:** The 2.4 GHz region is only available on chips with a 2.4 GHz radio path: LR2021, LR2022, LR1120, and LR1121. SX1261, SX1262, SX1268, and LR1110 are sub-GHz only.

## AGC command not recognized

**Symptom:** `agc auto` returns an error.

**Fix:** The `agc` command is available on LR20xx chips only (LR2021, LR2022). It is not supported on SX126x or LR11xx hardware.